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ESS 135W
Capstone Proposal

Title: Quantifying California Snow-to-Water Ratio Shifts from 2000 to 2050 to Understand Climate Change's Impact on Water Yields

Executive Summary:

Through this distinguished Fellowship, my study will quantify the shifts in the snow-to-rain ratio in California, specifically evaluating information from 2000-2050. The results would display the shifts and climate change impact, providing valuable information for water managers in the state, so that they may adjust their management strategies accordingly and effectively.

Statement of Need:

Climate change in California is expected to have a continuous impact on the state's water supply, increasing the likelihood of water shortages and drought (Sokolo et. al., 2019). Seasonal runoff from snowmelt has been identified as the state's main reservoir (Carle, 2019). However, research models have suggested that if greenhouse gas emissions are unchanging in the future, then western US states are likely to find themselves almost completely lacking snow in the next 35 to 60 years (Siirila-Woodburn et. al., 2021). This makes the ratio of snow-to-rain extremely important for scientists and managers to understand the changing implications of.

The entire state depends on precipitation input from the earth's hydrological cycle, as its supply is divided between three large consumption sectors (Mount & Hanak, 2019). As of May of 2019, the water supply in California is divided up into three main sections, environmental (50%), agricultural (40%), and urban (10%) (Mount & Hanak, 2019). With climate change, the allocations of the water supply will need to be adjusted in order for farmers, residents, habitats, etc. to have adequate water access. However, the implications of climate change are not well understood. Once the future effects are known, adaptive and dynamic management strategies can be put into place for sustainable water usage.

Adaptive management strategies for Northern California's water supply have already been explored (Georgakakos et. al., 2012). Water recycling (Sokolo et. al., 2019) and other strategies from built models (Georgakakos et. al., 2012) have been conceptualized and lightly applied. However, my study will provide managers with quantitative data of the entire state of California in order to determine the impact of climate change here on the snow-to-rain ratio. This will be crucial information allowing water managers in California to set up more sustainable and adaptive allocation strategies.

Project Goals and Description:

In this study, climate change impacts will be measured and quantified through the snow-to-rain ratio, which will be calculated through the Raster Calculator tool and conditional

statements. Temperature minimum and maximum datasets will be used to ensure accurate precipitation types are being represented on the map from the specific precipitation datasets. If precipitation occurs in a location where the temperature is less than or equal to 32°F, then that will be classified as snowfall. Anything else will be considered as rainfall. This will allow for an accurate representation of the expected changes in the snow-to-rain ratio in California.

Methodology and GIS Approach:

In order to carry out this study, I plan to use mapping and raster analysis techniques in ArcGIS Pro. The data layers I will use include precipitation data, temperature minimum and maximums, snowfall flux, and a California state boundary layer. These datasets, aside from the boundary layer and temperatures, were gathered as historical and future values. The future precipitation and snowfall flux data was gathered from the SSP3-7.0 projection, indicating business as usual emissions through 2050.

The datasets have all already been plugged into ArcGIS Pro and clipped to the study area of California. Precipitation and snowfall flux (both historical and future) datasets were uploaded into the ArcGIS Pro map by using the Make NetCDF Raster Layer tool, then clipped to the boundary (Figure 3; Figure 4; Figure 5; Figure 6).

There will be multiple minimum temperature (Figure 1) layers displayed according to and with other datasets within the same temporal range, which will be approximately 10-15 years each. The same extent applies to the maximum temperature (Figure 2) layers. Snowfall flux data will also be separated and calculated with the according layers in the same temporal period.

Snow-to-rain ratios for each temporal period will be calculated by dividing the snowfall flux by the precipitation rate, and multiplied by the density of water divided by the density of snow for each temporal period. This will be done using the Raster Calculator tool. The results of this study will be displayed in a comprehensive StoryMap, with maps for each temporal period(~4-5 maps).

Expected Outcomes and Broader Impacts:

It is expected that this study will show continuous decrease in snowfall and continuous increase in rainfall (Georgakakos et. al., 2012; Siirila-Woodburn et. al., 2021). The resulting map products will allow readers to see the shifts in the California snow-to-rain ratio in a clear and lucid manner for maximum interpretation by managers. Once the results of this study are determined and displayed, it will be easier for water managers to effectively plan for sustainable water usage every year. The timing of winter storms is another key factor and known to complicate water management, as it is typically inconsistent throughout each year (Dourado & Gustavo, 2024; Dourado & Viers, 2024). Therefore, in the future it would be beneficial to build off of this study from a smaller temporal scale, either monthly or daily. This would add to the ratio shifts that my study will provide. Once the actual impacts of climate change on the snow-to-rain ratios for each period, the timing factor will need to be emphasized in future efforts. This would then allow researchers to better understand the changing hydrological timing of

different kinds of precipitation, which can create distinctive and more dynamic sustainable water strategies.

References:

- Carle D (2019). Introduction to Water in California. University of California Press.
<https://books.google.com/books?hl=en&lr=&id=6golDQAAQBAJ&oi=fnd&pg=PR9&ots=8HbL8e-veT&sig=QVpoLe3Kj8CtKCgTdkJBnjinj5uw#v=onepage&q&f=false>
- Dourado F, Gustavo (2024). Balancing trade-offs for sustainable water resources management: reconciling climate change, hydropower, and environmental flows in the Central Sierra Nevada, California. <https://escholarship.org/uc/item/55h91051>.
- Dourado GF & Viers JH (2024). Reliability and resilience of environmental flows under uncertainty: reconsidering water year types and inconsistent flow requirements in California. *Environmental Research Letters* 19(9): 10.1088/1748-9326/ad6d80.
- Georgakakos KP, Graham NE, Cheng FY, Spencer C, Shamir E, Georgakakos AP, Yao H, Kistenmacher M (2012). Value of adaptive water resources management in northern California under climatic variability and change: Dynamic hydroclimatology. *Journal of Hydrology* (412-413):47-65. <https://doi.org/10.1016/j.jhydrol.2011.04.032>.
- Lange S, Büchner M (2021). ISIMIP3b bias-adjusted atmospheric climate input data. ISMIP Repository. 1.1: <https://doi.org/10.48364/ISIMIP.842396.1>.
- Mount JF, Hanak E (2019). Water Use in California. Managing Drought in a Changing Climate: Four Essential Reforms (1-6).
https://cwc.ca.gov/-/media/CWC-Website/Files/Documents/2019/06_June/June2019_Item_12_Attach_2_PPICFactSheets.pdf.
- Siirila-Woodburn ER, Rhoades AM, Hatchett BJ, Huning LS, Szinai J, Tague C, Nico PS, Feldman DR, Jones AD, Collins WD, Kaatz L (2021). A low-to-no snow future and its impacts on water resources in the western United States. *Nature Reviews Earth & Environment* (2):800-819. <https://doi.org/10.1038/s43017-021-00219-y>.
- Sokolo S, Godwin H, Cole BL (2019). Perspectives on the future of recycled water in California: results from interviews with water management professionals. *Journal of Environmental Planning and Management* 11(62):1908-1928.
<https://doi.org/10.1080/09640568.2018.1523051>.

Appendix:

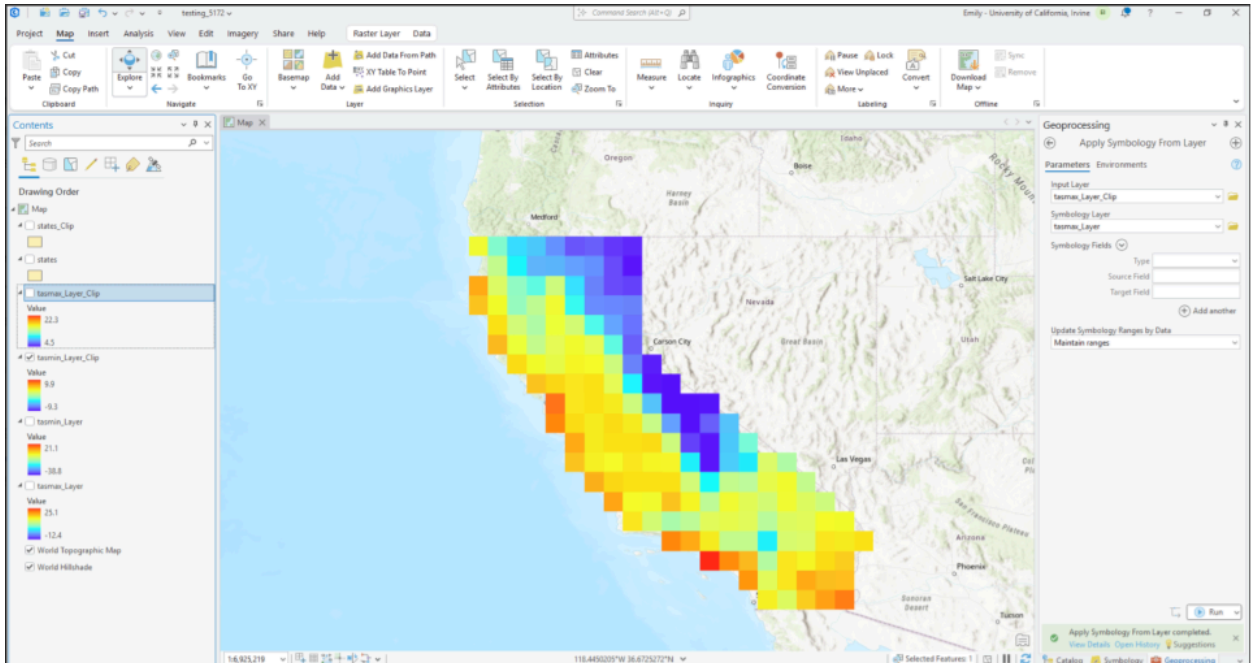


Figure 1. Shows the minimum temperature layer for the 2000-2014 temporal period. This layer was input to ArcGIS Pro through the Make a NetCDF Raster Layer tool, and then clipped to the study area extent. Lower values are shown in darker blues (cool colors) and higher temperature values are displayed in reds and warm tones.

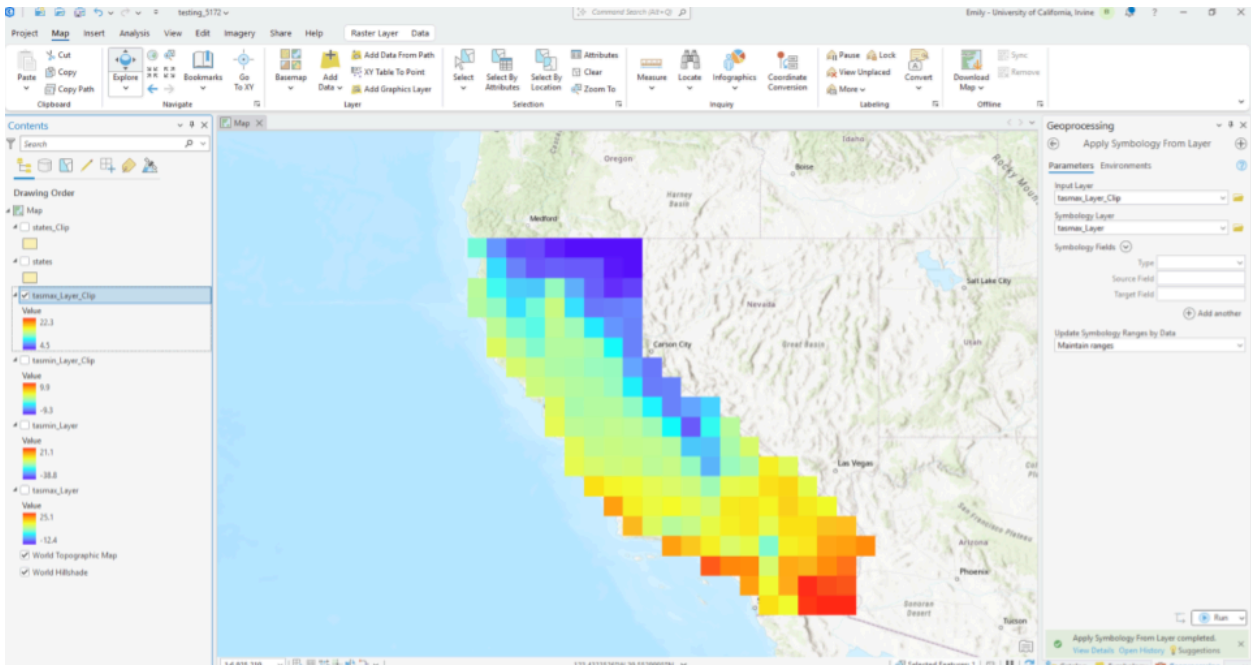


Figure 2. Shows the maximum temperature layer for the 2000-2014 temporal period. This layer was input to ArcGIS Pro through the Make a NetCDF Raster Layer tool, and then clipped to the California state boundary layer. Lower values are shown in darker blues (cool colors) and higher temperature values are displayed in reds and warm tones.

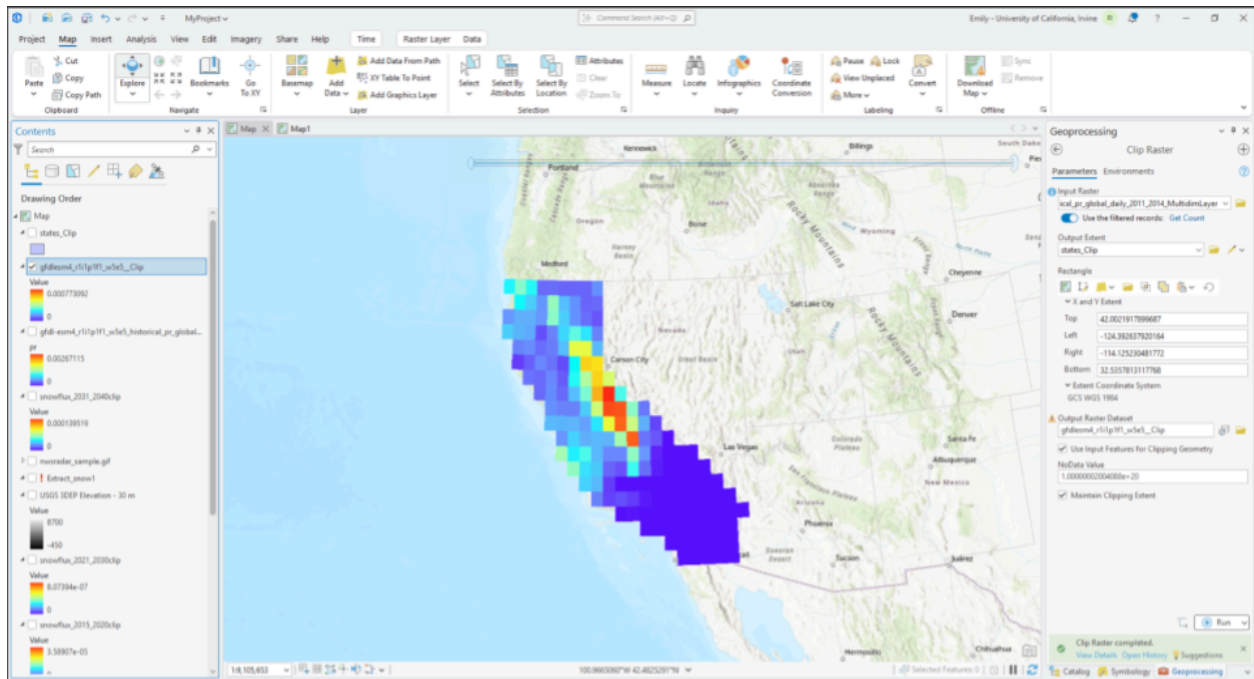


Figure 3. Shows the historical precipitation layer for the 2000-2014 temporal period. This layer was input to ArcGIS Pro through the Make a NetCDF Raster Layer tool, and then clipped to the California state boundary layer. Red and warm tones indicate higher precipitation values than the blue and cooler tones.

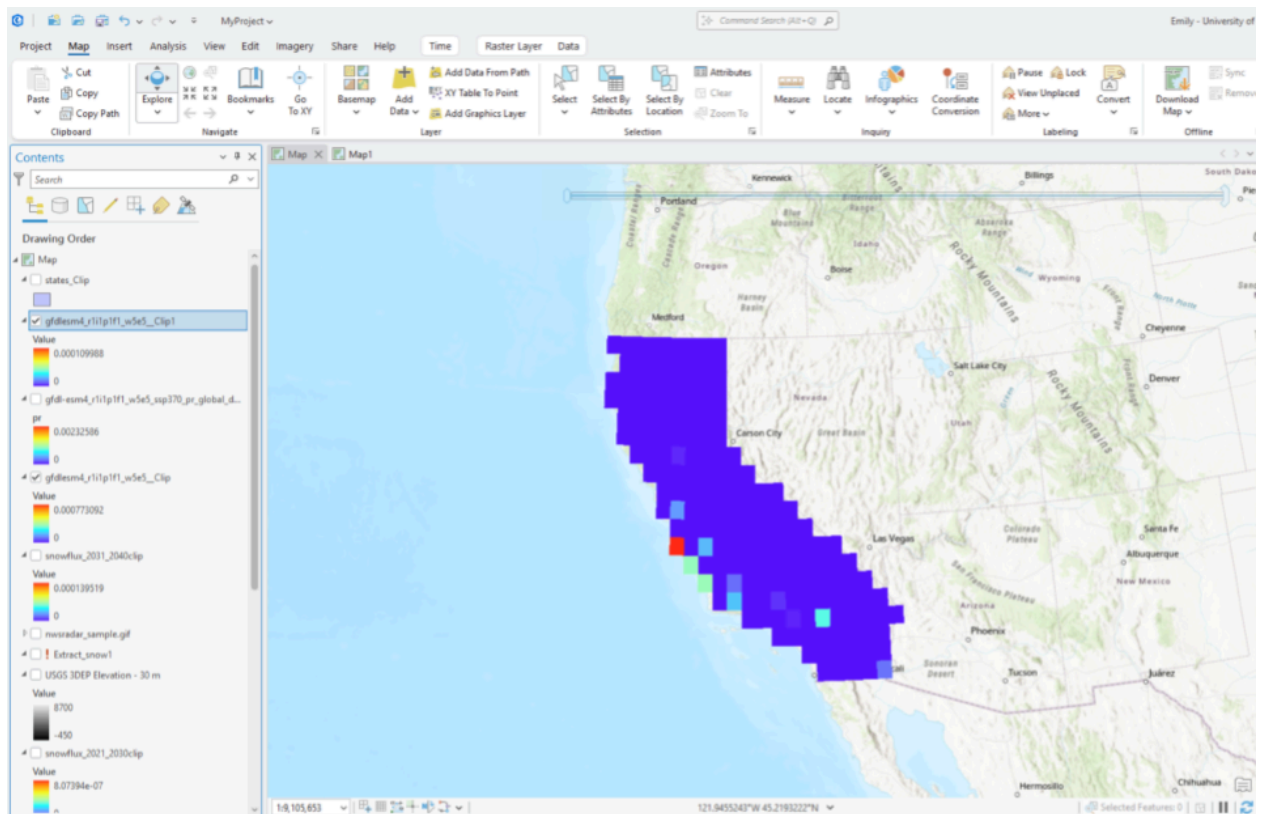


Figure 4. Shows the future precipitation data layer for the 2031-2040 temporal period. This layer was input to ArcGIS Pro through the Make a NetCDF Raster Layer tool, and then clipped to the study area.

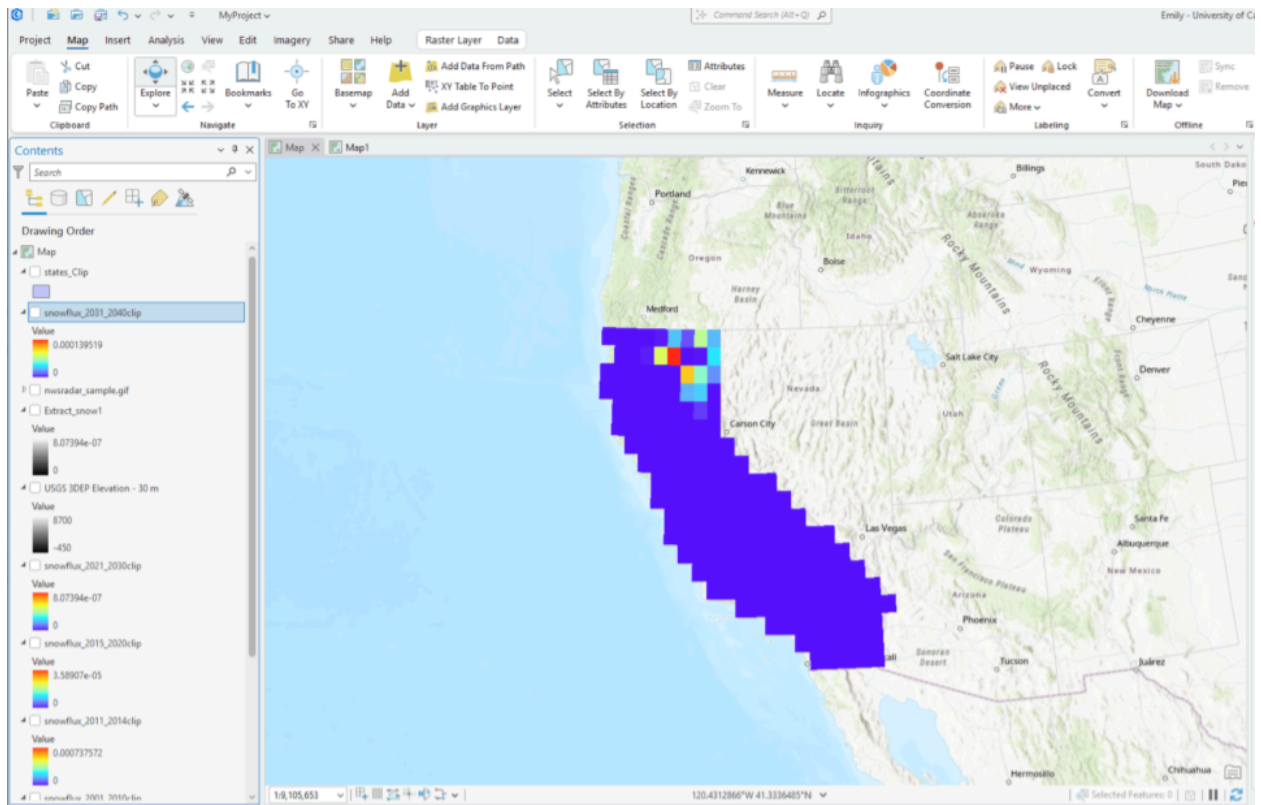


Figure 5. Shows the historical snowfall flux data layer for the 2000-2014 temporal period. This dataset was input to ArcGIS Pro through the Make a NetCDF Raster Layer tool, then clipped to the study area. Higher snowfall flux values are shown in reds and warmer tones while lower values are shown in blues and cooler tones.

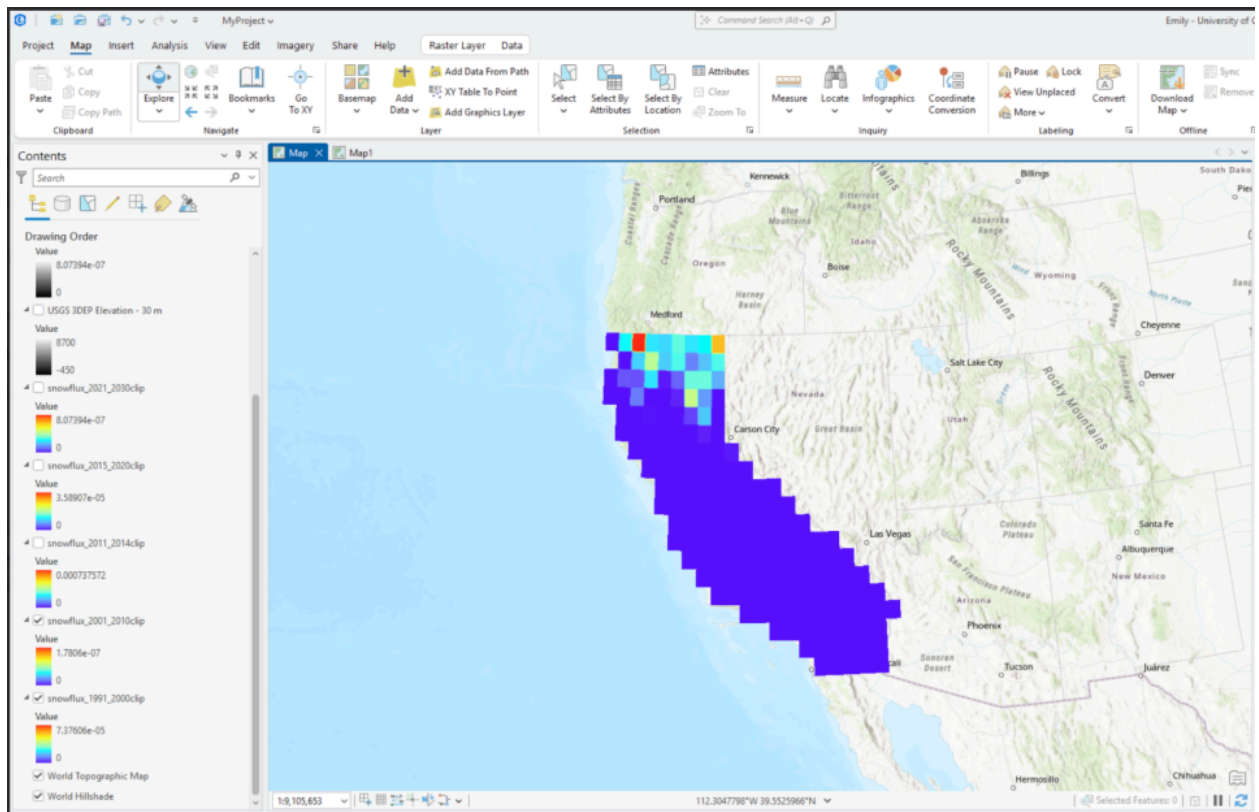


Figure 5. Shows the future snowfall flux data layer for the 2031-2040 temporal period. This dataset was input to ArcGIS Pro through the Make a NetCDF Raster Layer tool, then clipped to the California state boundary layer. Higher snowfall flux values are shown in red and warmer tones while lower values are shown in blues and cooler tones.